

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Scenario-Based Assessment

DURATION: 1.5 Hrs

1. A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns. Report how the RL framework can be applied by defining the possible states, actions, and reward functions for this scenario. **(5 Marks)**
2. A warehouse robot uses Reinforcement Learning to find the shortest path for delivering packages while avoiding obstacles. Sometimes the robot chooses longer paths due to insufficient exploration during training. Solve how the exploration vs exploitation trade-off affects the robot's navigation performance and suggest some improvements. **(5 Marks)**
3. A hospital is considering using Reinforcement Learning to optimize patient appointment scheduling in order to reduce waiting time and improve resource utilization. Demonstrate the suitability of RL for this scenario and discuss possible challenges such as delayed rewards and ethical considerations. **(5 Marks)**
4. A smart home system aims to reduce electricity consumption by automatically controlling lights, fans, and air conditioners based on user presence and weather conditions. Sketch a Reinforcement Learning model for this scenario by defining state space, action space, reward structure, and suitable learning algorithm.

(5 Marks)

5. An online gaming platform uses Reinforcement Learning to match players of similar skill levels to maintain engagement. Report how reward design can influence matchmaking quality with reward definition that might lead to biased or inefficient matching results. (5 Marks)

6. A financial trading system applies Reinforcement Learning to decide buy/sell actions based on market signals. Analyze how states, actions, and rewards can be defined in this context and evaluate the risks of delayed rewards in volatile markets.

(5 Marks)

7. A drone delivery company uses RL to optimize delivery routes while avoiding restricted airspace. Construct a suitable RL framework by defining states, actions, and rewards, and justify how penalties can ensure safe navigation. (5 Marks)

8. A university applies RL to personalize online learning paths for students. Examine how reward design can influence student engagement and defend the ethical considerations of adapting learning content dynamically. (5 Marks)

9. A traffic management authority deploys RL to control traffic lights at busy intersections. Illustrate how constraints such as emergency vehicle priority and pedestrian safety can be incorporated into the RL model and interpret the impact on congestion reduction. (5 Marks)

10. A manufacturing plant uses RL to schedule machines under resource limitations. Differentiate between traditional optimization methods and constrained RL approaches, and assess which method provides better adaptability in dynamic production environments. (5 Marks)

ANSWER

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1. Personalized Mobile Notifications

RL can learn when and what notification to send.

- **States:** User activity, time, location, previous notification response, app usage.
- **Actions:** Send notification, delay notification, or do not send.
- **Reward:** +10 for opening/clicking, +5 for engagement, -5 for ignoring, -10 for disabling notifications.

- The agent learns the best notification timing from user behavior.
- This improves engagement while reducing unwanted notifications.

2. Warehouse Robot – Exploration vs Exploitation

Exploration means trying new paths, while **exploitation** means selecting the currently best-known path.

- Too much exploitation may cause the robot to miss shorter paths.
- Too much exploration may cause unnecessary movements.
- Use **ϵ -greedy strategy** to balance both.
- Start with high ϵ and gradually reduce it.
- Experience replay and better reward design can further improve shortest-path learning.

3. Hospital Appointment Scheduling

RL is suitable because scheduling decisions affect future waiting times and resource usage.

- **State:** Patient queue, doctor availability, room availability.
- **Actions:** Assign appointment time, doctor, and room.
- **Reward:** Lower waiting time and better resource utilization.
- **Challenge:** Rewards may be delayed because the effect of a scheduling decision occurs later.
- Ethical issues include fairness, patient priority, privacy, and avoiding discrimination.

4. Smart Home Energy Management

An RL model can automatically control home appliances.

- **State:** User presence, temperature, weather, energy consumption.
- **Actions:** Turn lights/fans/AC on or off and adjust settings.
- **Reward:** Negative electricity consumption/cost and positive comfort.
- **Algorithm:** Q-Learning or Deep Q-Network (DQN).
- The agent learns energy-efficient control while maintaining user comfort.

5. RL-Based Game Matchmaking

The system uses RL to match players with suitable opponents.

- **State:** Player skill, experience, recent performance, waiting time.
- **Action:** Select a suitable opponent/team.
- **Good reward:** Higher reward for balanced and enjoyable matches.
- A reward based only on quick matchmaking may create unfair matches.
- Therefore, reward should consider **skill balance, player satisfaction, waiting time, and retention.**

6. RL Financial Trading

RL can learn trading decisions from market conditions.

- **State:** Stock price, trading volume, technical indicators, portfolio balance.
- **Actions:** Buy, sell, or hold.
- **Reward:** Profit after considering transaction costs and risk.
- Rewards may be delayed because profits/losses become clear later.
- Volatile markets can produce unstable rewards and high financial risk, so risk penalties and careful validation are required.

7. Drone Delivery

RL can optimize safe drone delivery routes.

- **State:** Drone location, battery level, weather, traffic, restricted zones.
- **Actions:** Move north/south/east/west, change altitude, wait.
- **Reward:** Positive reward for successful delivery and shorter routes.
- **Penalty:** Large negative reward for entering restricted airspace, collision, or low-battery risk.
- Safety penalties make dangerous actions less likely and encourage safe navigation.

8. Personalized Online Learning

RL can select suitable learning content for each student.

- **State:** Student performance, progress, difficulty level, engagement.
- **Actions:** Select topic, quiz, video, or difficulty level.
- **Reward:** Higher reward for learning improvement and engagement.
- Poor reward design may encourage only easy content instead of real learning.
- Ethical issues include privacy, fairness, bias, transparency, and avoiding inappropriate personalization.

9. RL Traffic Signal Control

RL can dynamically control traffic lights.

- **State:** Vehicle queue, traffic density, pedestrian crossing status.
- **Actions:** Change or maintain traffic signal phases.
- **Reward:** Reduce waiting time and congestion.
- **Constraints:** Emergency vehicles receive immediate priority, while minimum pedestrian crossing time must be maintained.
- Proper constraints reduce congestion while maintaining **safety and emergency response requirements**.

10. Machine Scheduling – Traditional vs Constrained RL

Traditional Optimization	Constrained RL
Uses predefined models	Learns from interaction
Works well for known conditions	Adapts to changing conditions
May require re-optimization	Can continuously adapt
Constraints are explicitly defined	Constraints can be incorporated into learning
Less flexible for dynamic environments	More suitable for dynamic environments

Conclusion: Traditional optimization is effective for stable production conditions, while **Constrained RL is more adaptable** when machine availability, demand, or resources change frequently.

Code of Conduct:

I ___G.Srivardhan___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Srivardhan

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

